

Operational Escalation Report —

OER-2025-002

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From: Margaret H. Ostrowski, Vice President, Operations — Cascadia Renewable Operations LLC

To: Chief Operating Officer; Chief Financial Officer; Board Audit & Risk Committee

cc: Chief Executive Officer; SVP Asset Management; General Counsel; VP Procurement; Director, Internal Audit

Subject: Western Region Operational Risk — Year-End FY2025 Assessment

Classification: Internal — Restricted Distribution

Executive Summary

This memorandum is a year-end assessment of operational risk in the Northwest Power Pool (NW-PowerPool) region. It is intended to inform the FY2026 capital plan and to put on the record a set of conditions that, in my judgment as VP of Operations, the company can no longer treat as routine.

Five NW-PowerPool sites — chiefly WIND-024, WIND-014, and WIND-003 — have generated a disproportionate share of FY2025 forced outages and emergency vendor spend. Across the year the region absorbed **\$8.9M** in emergency vendor invoices, against an internal preventive-scope equivalent that I estimate would have cost no more than \$2.3M. The Fall Overheating Sequence (Period B, October–November 2025) added 22,996 MWh of lost generation on top of Period A's 24,586 MWh. The corrective-action backlog stands at **31 open work orders**, the oldest dated December 2024 (CAR-2024-118). One management-of-change request — MOC-2025-022, governing the Siemens Gamesa OEM replacement threshold — has been pending engineering review for **nine months** and is now blocking a \$1.2M-per-unit decision on WIN-10131.

OER-2025-001 (September 2025) addressed the procurement symptom. This memorandum addresses the underlying asset and policy condition. The same vendor is being called repeatedly to the same site to perform the same emergency repair on aging assets, while a documented OEM-versus-internal-standard conflict prevents the engineering decisions that would break the cycle. The Board Audit & Risk Committee should be aware of the pattern before FY2026 budgeting.

I am requesting three Board-level decisions: a capital plan line item for aging-asset replacement at NW-PowerPool-WIND-024; formal renegotiation of the Siemens

Gamesa OEM service agreement; and asset-level retirement decisions for WIN-10130 and TRA-10561.

Section 1 — Pattern Description

NW-PowerPool generated 67% of all FY2025 forced outage MWh from approximately 18% of the company's installed capacity. Within NW-PowerPool, three sites account for the bulk of the impact:

Site	Forced outages FY25	Lost MWh	Emergency invoices
NW-PowerPool-WIND-O24	28+	~37,000	\$4.1M
NW-PowerPool-WIND-O14	21	~24,000	\$2.6M
NW-PowerPool-WIND-O03	14	~16,500	\$1.8M
NW-PowerPool, other	19	~11,500	\$0.4M
NW-PowerPool total	82+	~89,000	\$8.9M

The headline assets behind these numbers are also the assets that appear in our top-10 list of single-unit lost generation: WIN-10131 (17,239 MWh), WIN-10011 (14,411 MWh), SUB-10641 (13,362 MWh), WIN-10016 (12,589 MWh), WIN-10058 (11,606 MWh), WIN-10085 (8,200 MWh), TRA-10561 (7,900 MWh), and SUB-10629 (9,800 MWh). All eight sit on three sites.

Period B compounded Period A on the same equipment. WIN-10131 had three Apex dispatches and then took a fourth in November. TRA-10561 had a May vibration-excess outage (OIR-2025-007) and then a December overheating outage (OIR-2025-004) — same transformer, same vendor, same emergency dispatch pattern, seven months apart. RCA-2025-002 establishes that the December event on TRA-10561 was not a new failure mode; it was a continuation of an inadequately remediated May condition.

Section 2 — Quantified Business Impact

Full-year financial picture, NW-PowerPool:

Item	FY25 Actual	FY25 Plan	Variance
Revenue	\$121.1M	\$128.4M	-\$7.3M

Item	FY25 Actual	FY25 Plan	Variance
Maintenance cost (reported)	\$13.4M	\$11.2M	+\$2.2M
Emergency vendor invoices (Apex + Siemens Gamesa OEM service)	\$8.9M	\$1.5M	+\$7.4M
Net profit	\$103.5M	\$115.7M	-\$12.2M

The \$8.9M emergency spend breaks down by vendor as follows:

Vendor	Activations	Total invoiced
Apex Emergency Grid Solutions	14	\$5.3M
Siemens Gamesa OEM Service (via Apex subcontract)	8	\$2.6M
Siemens Gamesa OEM Service (direct warranty work)	6	\$1.0M
Total	28	\$8.9M

Asset-replacement quotes outstanding:

- WIN-10131 gearbox (Siemens Gamesa): \$1.2M
- WIN-10011 gearbox (Siemens Gamesa, draft quote): \$1.2M
- TRA-10561 transformer (Siemens Energy, replacement option): \$2.4M
- SUB-10641 substation breaker (ABB, retrofit option): \$0.9M
- **Replacement quotes pending decision: \$5.7M**

Deferred maintenance backlog:

- 31 open corrective action work orders across NW-PowerPool
- 17 are on the three headline sites (WIND-024, -014, -003)
- 9 carry dates older than 180 days
- Estimated catch-up cost at CPS rates: \$3.8M

Section 3 — Root Causes

Four converging root causes:

3.1 Aging assets with no replacement plan — RCA-2025-002, RCA-2025-003.

WIND-024 turbines (commissioned 2009–2011) are operating 4–6 years past their original mid-life refurbishment point. WIN-10131 and WIN-10011 are the worst-performing of the cohort. RCA-2025-002 identifies bearing-race fatigue consistent with end-of-design-life condition. RCA-2025-003 attributes the November SUB-10629 soc_drift event to a battery management system whose original installation manual specifies a 7-year service life — exceeded in 2024.

3.2 OEM-versus-internal-standard conflict — CAR-2025-002, MOC-2025-022.

Siemens Gamesa OEM Manual Rev. 4.2 mandates full component replacement at vibration thresholds 30% lower than internal Standard MS-2025-03 requires. The OEM threshold drives the \$1.2M-per-unit gearbox quotes. MOC-2025-022 has been pending engineering review since March 2025. CAR-2025-002 documents seven instances where the OEM threshold was applied because the override pathway was unavailable.

3.3 Procedural shortcut under fatigue — RCA-2025-004.

RCA-2025-004 establishes that the October 7 WIN-10085 overheating event (Period B) was preceded by a skipped CSM-7 oil-spectrometer step during the summer service window. The skip was documented as a fatigue-driven shortcut by a crew that had exceeded Fatigue Management Policy FMP-2's 60-hr/wk limit for three consecutive weeks. Wildfire-season access restrictions had compressed the service window.

3.4 Procurement framework rewarding outages — see OER-2025-001.

The emergency-procurement framework, taken with the OEM threshold and the deferred-PM backlog, creates a financial loop in which the most reliable way to get budget approved at NW-PowerPool is to fail catastrophically. OER-2025-001 documented this pattern at the procurement layer; this memorandum places it in the asset and policy layer.

Section 4 — Resources At Risk

The following assets are, in my engineering judgment, beyond the point at which continued emergency repair is the rational economic choice:

- **WIN-10130** (Siemens Gamesa, CAISO-WIND-026) — 14 forced outages, 20,494 MWh lost (highest single-asset lost generation in the fleet). Recommend retirement decision in Q1 FY2026.
 - **TRA-10561** (Siemens, NW-PowerPool-WIND-003) — two emergency repairs in seven months on the same transformer. RAM-2025-004 escalated the repair-versus-replace decision; no determination yet. Recommend retirement-or-replace decision in Q1 FY2026.
 - **WIN-10131** (Siemens Gamesa, NW-PowerPool-WIND-024) — Siemens Gamesa quote of \$1.2M for gearbox replacement is in active dispute pending MOC-2025-022.
 - **SUB-10641** (ABB, NW-PowerPool-WIND-014) — battery-management system past design life; retrofit quoted at \$0.9M.
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Section 5 — Recommended Executive Action

1. **Capital plan line item — Aging-asset replacement program at NW-PowerPool-WIND-024.** Authorize \$18M over FY26–FY27 for sequential replacement of the WIND-024 turbine cohort, beginning with WIN-10131 and WIN-10011. Replacement program economics: \$18M capital against forecasted emergency-cost avoidance of \$5–7M per year and recovered generation of approximately 35,000 MWh annually.
2. **Renegotiate the Siemens Gamesa OEM service agreement.** The current OEM service relationship is structured around component replacement, not condition-based maintenance. Direct procurement to open negotiations with the objective of a condition-based service tier with threshold flexibility aligned to internal Standard MS-2025-03. Close MOC-2025-022 in conjunction.
3. **Asset-level retirement decisions for WIN-10130 and TRA-10561.** Both assets have crossed the threshold at which continued repair is no longer economically defensible. Direct engineering to issue retirement-versus-replace determinations by 2026-02-28. RAM-2025-004 should be folded into this determination for TRA-10561.

Secondary actions (delegable, do not require Board approval):

1. Complete the Heartland Reliability Partners pilot under VRA-2025-004 (already approved in OER-2025-001 follow-through).
2. Close CAR-2025-002 with binding adoption of Standard MS-2025-03 thresholds.
3. Mandatory Fatigue Management Policy FMP-2 audit at NW-PowerPool, Q1 FY2026.

Section 6 — Decision Required

I request:

- Board Audit & Risk Committee sign-off on the \$18M capital line item at the February 2026 Board meeting.
- COO direction to procurement to open the Siemens Gamesa renegotiation by 2026-01-31.
- COO direction to engineering to issue retirement-versus-replace determinations on WIN-10130 and TRA-10561 by 2026-02-28.

A reply by 2026-01-15 would allow these items to be folded into the FY2026 operating plan ahead of Q1 close.

Submitted by:

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Cascadia Renewable Operations LLC

References:

- OER-2025-001 — Emergency Vendor Cost Escalation, Q3 FY2025
- RCA-2025-002 — Bearing-race fatigue, WIN-10131 / WIN-10011
- RCA-2025-003 — SUB-10629 soc_drift, November 2025
- RCA-2025-004 — CSM-7 oil-spectrometer skip, WIN-10085 October 2025
- CAR-2025-002 — OEM-versus-internal-standard threshold conflict
- MOC-2025-022 — OEM Threshold Override (pending — 9 months)
- CAR-2024-118 — Preventive Maintenance Cadence (open from Dec 2024)
- RAM-2025-004 — Repair-versus-replace, TRA-10561

Distribution: COO; CFO; Board Audit & Risk Committee; CEO; SVP Asset Management; General Counsel; VP Procurement; Director, Internal Audit. Hold on file: Compliance, Board Secretary.